

# POWER BI

## WEEK 1 TASK

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# 1. Introduction

This report summarizes the activities, learnings, and deliverables completed during Week 1 of the Power BI internship. The focus of this week was to understand the problem statement, select a suitable dataset, and plan the data strategy for building an interactive Power BI report.

## 2. Objectives of Week 1

- Understand the overall goal of the Power BI project.
- Select a publicly available dataset suitable for analysis.
- Perform initial data exploration and identify key fields.
- Plan the data model (tables, relationships, primary/foreign keys).
- Document the process with screenshots and explanations.

## 3. Dataset Selection

**Dataset Name:** [e.g., “Sales Data”, “HR Attrition”, “Credit Card Transactions”]

**Source:** [Kaggle / GitHub / Government Portal / Company-provided]

**Why this dataset?:**

Contains multiple dimensions (e.g., time, product, customer).

Suitable for building KPIs and interactive visuals.

Clean enough for initial modeling, but still needs transformation.[github+1](#)

## **4. Data Strategy and Planning**

### **Data Cleaning Plan**

- Remove duplicates and nulls in key columns.
- Standardize date formats.
- Create calculated columns (e.g., Month, Quarter, Year).
- Handle outliers if present

## **5. Tasks Completed**

- Installed and set up Power BI Desktop.
- Imported the selected dataset into Power BI.
- Explored data using Power Query Editor.
- Performed basic cleaning (removed blanks, fixed data types).
- Created a simple data model with 2–3 tables and relationships.
- Documented each step with screenshots.

# SCREENSHOT 1 :

## Title: Data Import into Power Query Editor

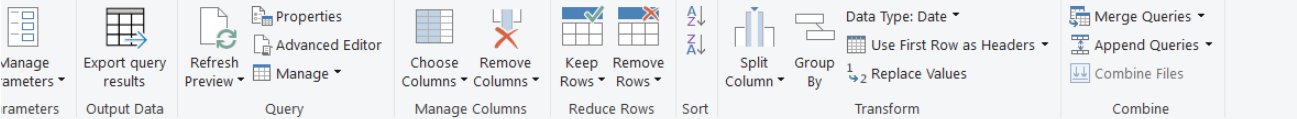
The Superstore dataset was successfully imported into Power Query Editor for data cleaning and transformation.

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      |
|--------|----------------|------------|------------|----------------|
| 1      | US-2023-103800 | 01-03-2023 | 01-07-2023 | Standard Class |
| 2      | US-2023-112326 | 01-04-2023 | 01-08-2023 | Standard Class |
| 3      | US-2023-112326 | 01-04-2023 | 01-08-2023 | Standard Class |
| 4      | US-2023-112326 | 01-04-2023 | 01-08-2023 | Standard Class |
| 5      | US-2023-141817 | 01-05-2023 | 01-12-2023 | Standard Class |
| 6      | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 7      | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 8      | US-2023-106054 | 01-06-2023 | 01-07-2023 | First Class    |
| 9      | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 10     | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 11     | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 12     | US-2023-130813 | 01-06-2023 | 01-08-2023 | Second Class   |
| 13     | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 14     | US-2023-167199 | 01-06-2023 | 01-10-2023 | Standard Class |
| 15     | US-2023-105417 | 01-07-2023 | 01-12-2023 | Standard Class |
| 16     | US-2023-105417 | 01-07-2023 | 01-12-2023 | Standard Class |
| 17     | US-2023-135405 | 01-09-2023 | 1/13/2023  | Standard Class |
| 18     | US-2023-135405 | 01-09-2023 | 1/13/2023  | Standard Class |
| 19     | US-2023-149020 | 01-10-2023 | 1/15/2023  | Standard Class |

# SCREENSHOT 2 :

## Title: Text Cleaning and Data Transformation

The Superstore dataset was successfully imported into Power Query Editor. The dataset contains multiple columns such as Row ID, Order ID, Order Date, Ship Date, and Ship Mode. The imported records are displayed in rows for further data cleaning and transformation.



The screenshot displays the Power Query Editor interface. At the top, a ribbon contains various data transformation options categorized into 'Manage Parameters', 'Output Data', 'Query', 'Manage Columns', 'Reduce Rows', 'Sort', 'Transform', and 'Combine'. The formula bar shows the query name 'Changed Type with Locale' and the transformation applied: 'Table.TransformColumnTypes'. The main area displays a table with the following data:

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      |
|--------|----------------|------------|------------|----------------|
| Valid  | 100%           | Valid      | 100%       | Valid          |
| Error  | 0%             | Error      | 0%         | Error          |
| Empty  | 0%             | Empty      | 0%         | Empty          |
| 1      | US-2023-103800 | 03-01-2023 | 07-01-2023 | Standard Class |
| 2      | US-2023-112326 | 04-01-2023 | 08-01-2023 | Standard Class |
| 3      | US-2023-112326 | 04-01-2023 | 08-01-2023 | Standard Class |
| 4      | US-2023-112326 | 04-01-2023 | 08-01-2023 | Standard Class |
| 5      | US-2023-141817 | 05-01-2023 | 12-01-2023 | Standard Class |
| 6      | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 7      | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 8      | US-2023-106054 | 06-01-2023 | 07-01-2023 | First Class    |
| 9      | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 10     | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 11     | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 12     | US-2023-130813 | 06-01-2023 | 08-01-2023 | Second Class   |
| 13     | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 14     | US-2023-167199 | 06-01-2023 | 10-01-2023 | Standard Class |
| 15     | US-2023-105417 | 07-01-2023 | 12-01-2023 | Standard Class |
| 16     | US-2023-105417 | 07-01-2023 | 12-01-2023 | Standard Class |
| 17     | US-2023-135405 | 09-01-2023 | 13-01-2023 | Standard Class |
| 18     | US-2023-135405 | 09-01-2023 | 13-01-2023 | Standard Class |
| 19     | US-2023-149020 | 10-01-2023 | 15-01-2023 | Standard Class |

# SCREENSHOT 3 :

## Topic:

## Text Cleaning and Data Transformation

The text columns in the Superstore dataset were cleaned using Power Query. Text trimming was applied to remove unnecessary spaces from fields such as Sub-Category and Segment. The cleaned data was then checked to ensure that the values were properly formatted.

Manage Parameters

Export query results

Refresh Preview

Advanced Editor

Manage

Query

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Use First Row as Headers

Replace Values

Transform

Merge Queries

Append Queries

Combine Files

Combine

Properties

Parameters

Output Data

Query

Manage Columns

Reduce Rows

Sort

Transform

Combine

Properties

Parameters

Output Data

Query

Manage Columns

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Properties

Parameters

Output Data

Query

Manage Columns

Reduce Rows

<

# SCREENSHOT 4 :

**Title:**  
**Order Date Transformation**

The Order Date column was separated into individual date components such as **Year, Quarter, Month, Day, and Day of Week** using Power Query. This transformation makes the date information easier to analyze and use for further data analysis.

fx

= Table.AddColumn(#"Inserted Day", "Day Name", each Date.DayOfWeekName([Order Date]), type text)

| <div><div></div><div>1<sup>2</sup>3</div><div>Quarter</div></div>                                               | <div><div></div><div>1<sup>2</sup>3</div><div>Month</div></div>                                                 | <div><div></div><div>1<sup>2</sup>3</div><div>Day</div></div>                                                   | <div><div></div><div>A<sup>B</sup>C</div><div>Day Name</div></div>                                              |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <div><div>100%</div><div>0%</div><div>0%</div><div><div>Valid</div><div>Error</div><div>Empty</div></div></div> | <div><div>100%</div><div>0%</div><div>0%</div><div><div>Valid</div><div>Error</div><div>Empty</div></div></div> | <div><div>100%</div><div>0%</div><div>0%</div><div><div>Valid</div><div>Error</div><div>Empty</div></div></div> | <div><div>100%</div><div>0%</div><div>0%</div><div><div>Valid</div><div>Error</div><div>Empty</div></div></div> |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 3 Tuesday                                                                                                       |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 4 Wednesday                                                                                                     |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 4 Wednesday                                                                                                     |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 4 Wednesday                                                                                                     |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 5 Thursday                                                                                                      |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 6 Friday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 7 Saturday                                                                                                      |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 7 Saturday                                                                                                      |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 9 Monday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 9 Monday                                                                                                        |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 10 Tuesday                                                                                                      |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 10 Tuesday                                                                                                      |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 11 Wednesday                                                                                                    |
| 2023                                                                                                            | 1                                                                                                               | 1                                                                                                               | 13 Friday                                                                                                       |

## **CONCLUSION :**

The Superstore dataset was successfully imported and prepared using Power Query. Data cleaning and date transformations were performed to improve data quality and make the dataset suitable for further analysis. The transformed data is now ready for analysis and visualization in Power BI.